



A Project Report

KYAU Tuition App

— Dual-Role Academic Management System

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Department of Computer Science & Engineering

School of Science and Engineering

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KYAU Tuition App

Version 2.0 — Dual-Role Academic Management System

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Signature of Supervisor

Date: _____

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List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
APK	Android Package Kit
BSc	Bachelor of Science
CSE	Computer Science and Engineering
CSS	Cascading Style Sheets
DB	Database
GPA	Grade Point Average
HTML	Hyper-Text Markup Language
HOD	Head of Department
JS	JavaScript
JSON	JavaScript Object Notation
JSX	JavaScript XML
KAI	KYAU Artificial Intelligence (the in-app chatbot)
KYAU	Khwaja Yunus Ali University
LLM	Large Language Model
MAD	Mobile Application Development
MVC	Model-View-Controller
NoSQL	Not Only SQL
PWA	Progressive Web Application
REST	Representational State Transfer
SDK	Software Development Kit

Abbreviation	Full Form
SPA	Single-Page Application
UI	User Interface
UX	User Experience

Abstract

KYAU Tuition App (version 2.0) is an Android-compatible, web-based academic management system developed specifically for the students and faculty of the Computer Science and Engineering (CSE) department at Khwaja Yunus Ali University (KYAU). The application addresses the growing need for a centralised, role-aware digital platform that simplifies the day-to-day academic interactions between students and teachers within a university environment. The system offers two distinct portals — one for students and one for teachers — each equipped with tools tailored to their respective academic roles. Students can browse all eight semesters of the BSc CSE curriculum, submit and track assignments, view graded results with computed GPA, check their weekly class timetable, read official departmental notices, and interact with *KAI*, an AI-powered tutoring assistant built on Anthropic’s Claude language model. Teachers, on the other hand, can monitor student performance and attendance, create and manage assignments, grade submitted work, maintain a personalised class schedule, conduct real-time messaging with students, and use *KAI* as an intelligent teaching assistant. The application is built using React 18 and Vite on the frontend, with Firebase providing authentication and real-time database capabilities. The *KAI* chatbot is powered through direct API calls to Anthropic’s Claude Sonnet model. The entire application is packaged as a Progressive Web Application (PWA) and is convertible into a native Android APK using Capacitor, making it suitable for deployment on mobile devices without requiring a dedicated backend server build. This report presents a thorough account of the project’s background, design rationale, system architecture, implementation details, testing outcomes, and future improvement possibilities.

Chapter 1

Introduction

1.1 Background

Higher education institutions in Bangladesh increasingly depend on digital tools to manage and streamline their academic operations. However, many universities — particularly those at the regional level — still rely heavily on paper-based notice boards, informal messaging apps, and disconnected spreadsheets for managing class schedules, assignments, and student-teacher communication. This creates an inefficiency that affects both academic performance tracking and day-to-day departmental coordination.

Khwaja Yunus Ali University (KYAU) is a private university located in Enayetpur, Sirajganj, Bangladesh. Its CSE department runs a four-year BSc programme spanning eight semesters with a total of 150 credit hours. With students and faculty members spread across multiple batches and course sections, the absence of a unified digital management system becomes a genuine academic bottleneck.

The KYAU Tuition App was conceived in this context — as a purpose-built academic application for the CSE department that brings together the core functions of class management, assignment tracking, grading, scheduling, and intelligent academic assistance into a single cross-platform interface. Figure 1.1 illustrates the high-level interaction between the application’s main actors and services.

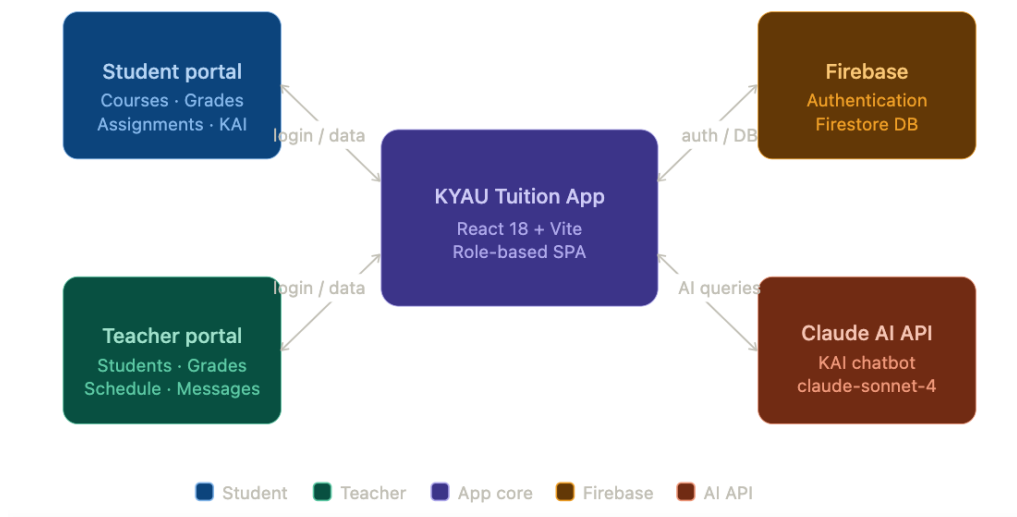


Figure 1.1: System overview: Student portal and Teacher portal interact through the app, which connects to Firebase (authentication and real-time database) and the Anthropic Claude AI API.

1.2 Problem Statement

The CSE department at KYAU currently lacks a dedicated digital platform that connects students and teachers in a meaningful academic workflow. Specifically, the following pain points motivated the development of this application:

- Students have no central place to view their assignments, submission status, and grades together.
- Class timetables are shared informally and are often outdated or inaccessible to students who miss notice board updates.
- Teachers must manage assignments and track student submissions through fragmented tools.
- There is no structured channel for one-to-one academic messaging between teachers and students within the university system.
- Students lack immediate academic assistance outside of class hours, particularly for CSE-related subject queries.

1.3 Motivation

The primary motivation behind this project was to build something genuinely useful for our own department — a tool that we, as final-year CSE students, would want to use every day. The integration of an AI tutoring assistant (KAI, powered by Claude AI) was particularly exciting because it represents the intersection of modern large language model technology and practical educational support.

Beyond personal relevance, the project was also motivated by the opportunity to apply the full stack of skills acquired during the Mobile Application Development course — from React component architecture and state management to Firebase integration and mobile packaging with Capacitor.

1.4 Objectives

The specific objectives of the KYAU Tuition App project are:

1. To design and implement a dual-role (student and teacher) academic management application for the KYAU CSE department.
2. To provide students with seamless access to course information, assignment management, grade tracking, timetable viewing, and official notices.
3. To equip teachers with tools for creating and grading assignments, managing class schedules, monitoring student performance, and communicating with students.
4. To integrate an AI-powered tutoring assistant (KAI) that provides context-aware academic support for both students and teachers.
5. To support both web and Android platforms through a single codebase using React and Capacitor.
6. To implement a real-time notification system that keeps both user groups informed of academic events.

1.5 Target Users

The application is designed for two primary user groups within the KYAU CSE department:

Students: Undergraduate BSc CSE students from any batch who need to manage their coursework, check grades, stay informed about class schedules and departmental notices, and seek academic help.

Teachers and Faculty: Lecturers, assistant professors, and the Head of Department who need to post assignments, grade submissions, track student attendance and performance, schedule classes, and communicate with students.

Chapter 2

Related Work

2.1 Existing Applications

A number of academic management and e-learning applications exist in the market today. The most prominent ones relevant to this project are:

Google Classroom: A widely-used platform by Google that allows teachers to create classes, post assignments, and grade submissions. It integrates seamlessly with Google Drive and Google Meet.

Microsoft Teams for Education: An all-in-one collaboration platform that includes assignment management, video conferencing, and file sharing for educational institutions.

Blackboard Learn: A comprehensive learning management system (LMS) used by many universities worldwide, offering course management, assessments, and reporting features.

Moodle: An open-source LMS that universities can host independently. It provides extensive plugin support and customisation options.

Canvas LMS: A cloud-based platform that supports course building, assignment management, grading, and student-teacher communication.

2.2 Comparison of Features

Table 2.1: Feature Comparison with Existing Applications

Feature	G. Class- room	MS Teams	Moodle	Canvas	KYAU App
University-specific data	No	No	No	No	Yes
Dual-role portal	Yes	Yes	Yes	Yes	Yes
AI tutor integration	No	Limited	No	No	Yes (KAI)
Offline-capable APK	No	Yes	No	No	Yes
8-semester course browser	No	No	No	No	Yes
Class timetable	No	Yes	Partial	Partial	Yes
Real-time messaging	No	Yes	Plugin	Yes	Yes
GPA computation	No	No	Plugin	Yes	Yes
Zero setup for students	Yes	No	No	No	Yes
Departmental notices	No	Yes	Yes	Yes	Yes

2.3 Limitations of Existing Solutions

While each of the above platforms is powerful in its own right, they share several limitations when considered for the specific context of a Bangladeshi regional university such as KYAU:

- None of them contain KYAU-specific curriculum data, course codes, faculty information, or departmental structure.
- Most require institutional licensing, IT infrastructure, or third-party authentication that may not be readily available.
- Their complexity and generalised design can be overwhelming for students and faculty who need straightforward, focused tools.
- None integrate a conversational AI tutor that is aware of the university’s own faculty,

courses, and academic context.

- Most are not easily convertible to standalone Android APKs for offline or low-bandwidth environments.

2.4 Our Improvements

KYAU Tuition App addresses each of these limitations directly:

- It embeds real KYAU CSE data — all eight semesters of courses with accurate credit allocations, faculty names and titles, class schedules, and departmental notices.
- It provides a clean, role-based interface that is purpose-built for KYAU students and teachers, with no unnecessary complexity.
- The KAI AI tutor knows the university’s faculty, course catalogue, department contact, and fee structure, making it contextually relevant to KYAU students.
- The entire application can be packaged as an Android APK using Capacitor, making it deployable as a native mobile app without requiring a separate mobile codebase.
- It is built entirely with open-source technologies (React, Vite, Firebase) and requires no institutional licensing.

Chapter 3

Tools and Technologies Used

3.1 Frontend Technologies

React 18 is the core JavaScript library used for building the user interface. React’s component-based architecture allowed us to build reusable UI elements — such as navigation bars, assignment cards, timetable rows, and grade tables — that are shared across both the student and teacher portals. React’s state management (via `useState` and `useEffect` hooks) handles the dynamic interactions within the application, such as switching between tabs, filtering assignments, and updating chat messages.

Vite serves as the build tool and development server. Vite’s extremely fast hot module replacement (HMR) made the development cycle significantly more productive compared to older bundlers like Webpack. The `npm run dev` command serves the application at `localhost:5173` during development.

Pure CSS was used for all styling. Rather than adopting a utility framework like Tailwind CSS, the team wrote custom CSS using CSS custom properties (variables) to maintain a consistent design token system across both portals. This kept the bundle size small and gave us complete control over the visual design.

JSX (JavaScript XML) is the syntax extension used within React components to define component structure in a readable, HTML-like format.

3.2 Backend and Authentication

Firebase provides the backend infrastructure for the application. Two Firebase services are used:

- **Firebase Authentication** — Handles user login with email and password for both students and teachers.
- **Cloud Firestore** — A real-time NoSQL database used to store and sync chat messages, notifications, and user session data across devices.

3.3 AI Integration

Anthropic Claude API powers the KAI (KYAU Artificial Intelligence) chatbot. The application makes direct API calls to Anthropic's `/v1/messages` endpoint using the `claude-sonnet-4-20250514` model. Each call constructs a context-aware system prompt that includes the university's faculty list, course catalogue, and user-specific information so that KAI can give relevant, personalised responses to both students and teachers.

3.4 Mobile Packaging

Capacitor is used to convert the React web application into a native Android APK. After running `npm run build`, the compiled `dist/` directory is synced with Capacitor and opened in Android Studio, where a signed APK can be generated for distribution.

3.5 Development Tools

- **Visual Studio Code** — Primary code editor used by all team members.
- **Git & GitHub** — Version control and collaborative code management.
- **Node.js & npm** — JavaScript runtime and package manager.
- **Android Studio** — Used for building and testing the Android APK.
- **Chrome DevTools** — Used for real-time debugging and responsive design testing.

Table 3.1: Summary of Technologies Used

Category	Technology	Version	Purpose
Frontend Framework	React	18	UI components
Build Tool	Vite	Latest	Dev server, bundling
Styling	Pure CSS	—	Design system
Backend/Auth	Firebase	10.x	Auth + Firestore DB
AI API	Anthropic Claude	Sonnet 4	KAI chatbot
Mobile Packaging	Capacitor	Latest	Android APK
Runtime	Node.js / npm	LTS	Package management
Version Control	Git / GitHub	—	Collaboration

Chapter 4

System Analysis and Design

4.1 System Overview

KYAU Tuition App follows a Single-Page Application (SPA) architecture. Once loaded in the browser (or as an APK on Android), the application manages all navigation internally through React’s state, without performing full page reloads. This results in a fast, app-like feel even on lower-end devices.

The application is structured around two roles: **student** and **teacher**. After authentication, the system reads the user’s role and renders the appropriate portal shell. Both portals share common components — the login page, notification dropdown, notification page, and the KAI chat page — but display role-appropriate content within them.

4.2 Architecture

The overall architecture follows a five-layered pattern as shown in Figure 4.1:

1. **Presentation Layer** — React components render the UI for both portals.
2. **Application/Logic Layer** — Custom hooks (`useAuth`, `useNotifications`) manage authentication and notification state.
3. **Data Layer** — Static university data (`kyauData.js`) and Firebase Firestore for dynamic real-time data.
4. **AI Layer** — KAI communicates with Anthropic’s Claude API with a context-rich system prompt on each call.

5. **Mobile Layer** — Capacitor bridges the React web app to Android, enabling APK distribution.

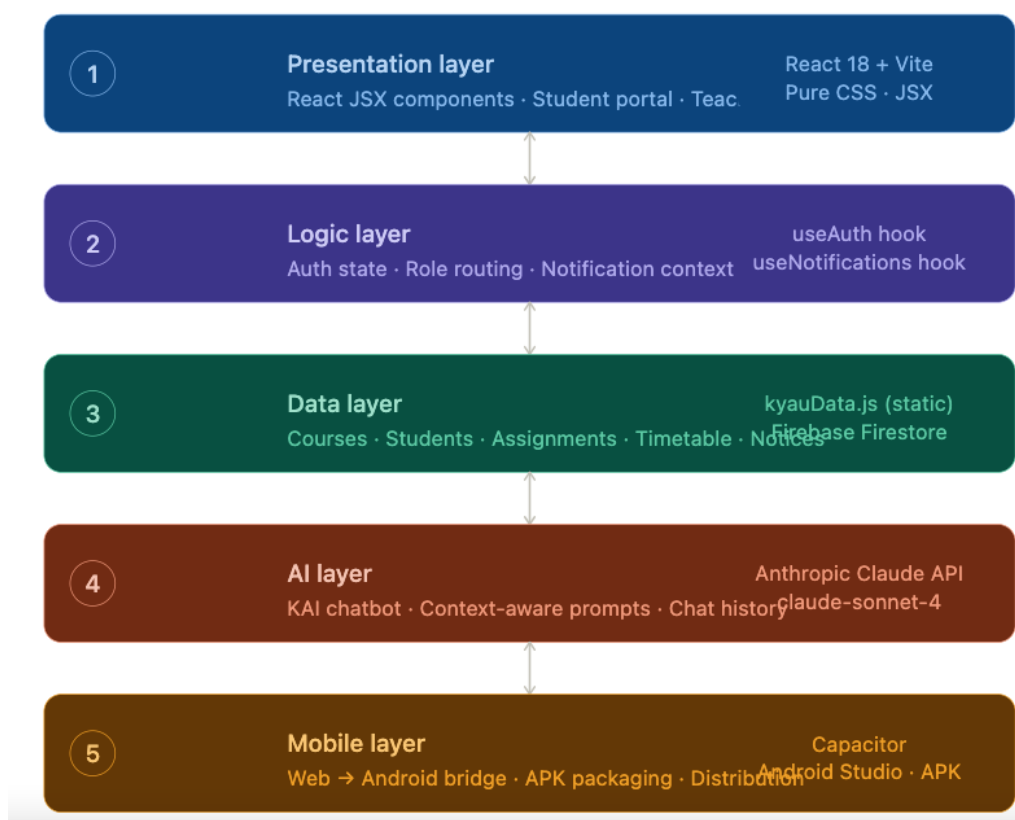


Figure 4.1: Five-layer application architecture: Presentation → Logic → Data → AI → Mobile.

4.3 Module Design

4.3.1 Authentication Module

The `AuthProvider` context wraps the entire application. Role determination is done from the email domain — teacher accounts use `@teacher.kyau.edu.bd` and student accounts use `@student.kyau.edu.bd`.

Demo credentials are available for three student accounts (`rafiq`, `sadia`, `jamil`) and three teacher accounts (`shamrat`, `rokeya`, `raihaan`).

4.3.2 Navigation Module

The `Shell` component renders the application frame with a responsive sidebar that collapses into a drawer on mobile (screen width ≤ 640 px). All navigation is state-driven

via the `navigate(id)` function — no URL-based router is used, making it ideal for APK packaging.

4.3.3 Student Portal Modules

Student Dashboard: Displays a personalised welcome card, stat tiles (Courses, Pending, Today's Classes, Attendance), today's timetable preview, current semester courses with progress bars, and the latest departmental notice.

Courses Module: Presents all eight semesters of the BSc CSE programme in a tab interface, with course code, name, credit count, and Theory/Lab type for each course across 150 total credits.

Assignments Module: Lists assignments with title, course, due date, and status (Pending, Submitted, Late, Graded). Tab-based filtering and context-sensitive action buttons are provided.

Grades Module: Shows all graded submissions in a table with GPA computed automatically using the 4.0-scale grade-point mapping.

Timetable Module: Shows the weekly class schedule (Sunday to Thursday) with time, course, teacher, room, and a Join button per class.

Notices Module: Displays official departmental announcements tagged by type (Exam, Event, Academic) with dates.

4.3.4 Teacher Portal Modules

Teacher Dashboard: Displays a welcome card, stat tiles (Students, To Grade, Active Tasks, Rating), today's schedule, pending submissions, and a student attendance overview.

Students Module: Searchable card grid of all enrolled students with GPA, attendance progress bars, and a detailed profile view per student.

Assignments Module: Three-view interface — list view, create view (form with course dropdown and due date), and grade view (inline grade input per submission).

Schedule Module: Teacher's personal weekly schedule filtered from the full department

timetable, with a toggle to view the entire department schedule.

Messages Module: Two-panel real-time chat — student list on the left, conversation history on the right.

4.3.5 KAI AI Chatbot Module

The KAI page adapts its persona and subject chips based on the user’s role. For students it acts as a friendly academic tutor; for teachers it acts as a professional teaching assistant. Each call sends the last 20 messages as history along with a university-context system prompt.

4.3.6 Notification Module

A bell icon in the top bar shows unread count as a badge. Clicking opens a dropdown preview; “View All” navigates to the full Notifications page.

4.4 Data Model

Table 4.1: Core Data Entities

Entity	Key Fields
SEMESTERS	label, full (name), courses (array)
Course	code, name, cr (credits), icon, color, lab (boolean)
FACULTY	name, title, email, phone
STUDENTS	id, name, email, year, gpa, attendance, color, avatar
TIMETABLE	day, time, course, teacher, room
ASSIGNMENTS_DATA	id, title, course, dueDate, status, submissions, total
STUDENT_SUBMISSIONS	id, studentId, name, file, submittedAt, status, grade
NOTICES	id, title, body, date, type, icon
MESSAGES_DATA	id, from, fromRole, to, toRole, text, time, date

4.5 User Interface Design Principles

The UI uses a card-based design language with role-specific colour palettes — blue (#1e50a0) for students and teal-green for teachers. Key decisions include role-based theming, responsive sidebar behaviour, horizontal progress bars for attendance and GPA visualisation, and colour-coded status pill badges for assignment states.

Chapter 5

Implementation

5.1 Project Structure

```
kyau-full/  
  src/  
    components/  
      shared/  
        LoginPage.jsx      -- Email/password login form  
        ChatPage.jsx       -- KAI AI chatbot (dual-role)  
        NotificationsPage.jsx -- Full notifications list  
        NotifDropdown.jsx  -- Bell icon dropdown preview  
        ToastContainer.jsx  -- Toast notification renderer  
      student/  
        StudentDashboard.jsx -- Student home screen  
        StudentCourses.jsx   -- 8-semester course browser  
        StudentAssignments.jsx -- Assignment tracking  
        StudentGrades.jsx    -- Graded work and GPA  
        StudentTimetable.jsx -- Weekly class schedule  
        StudentNotices.jsx   -- Departmental notices  
      teacher/  
        TeacherDashboard.jsx -- Teacher home screen  
        TeacherAssignments.jsx -- Create and grade work  
        TeacherStudents.jsx   -- Student list and profiles  
        TeacherSchedule.jsx   -- Class schedule management
```

```
TeacherMessages.jsx    -- Real-time student messaging
hooks/
  useAuth.jsx           -- Auth context and hook
  useNotifications.jsx  -- Notification context and hook
data/
  kyauData.js           -- Static university data
App.jsx                -- Root shell and routing
README.md
```

Listing 5.1: Project Directory Structure

5.2 Login / Home Page

Figure 5.1 shows the application’s home (login) page where users enter their KYAU email and password to access their respective portal.

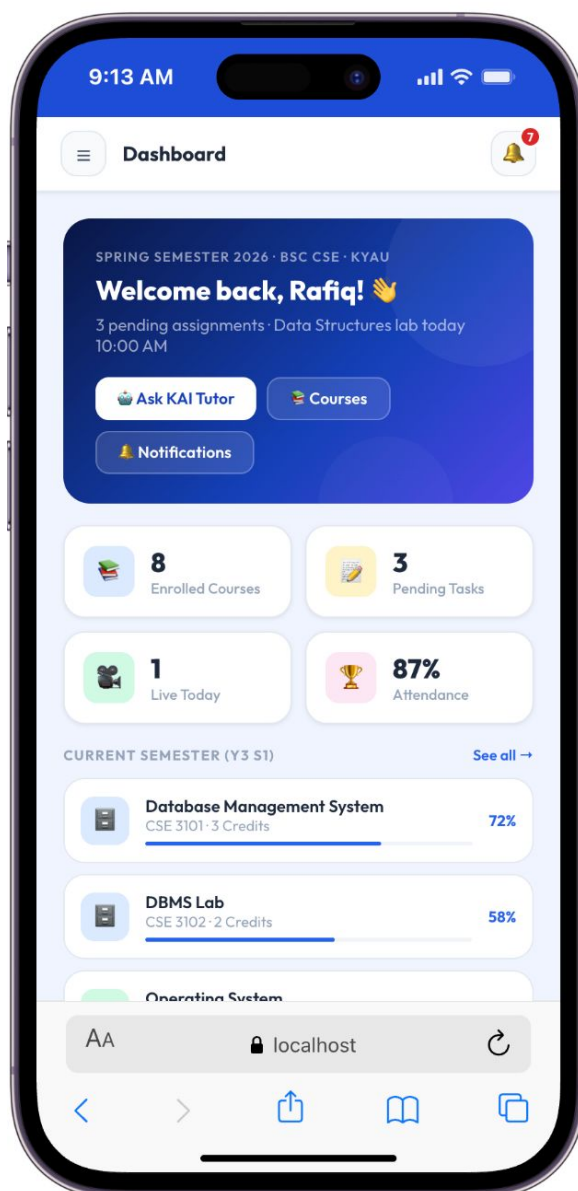


Figure 5.1: Application home page — users sign in with their KYAU email address; the role (student or teacher) is detected automatically from the email domain suffix.

5.3 Student Dashboard

After a successful login, students land on the Dashboard shown in Figure 5.2. The page greets the student by name and presents a quick overview of their academic standing for the current semester.

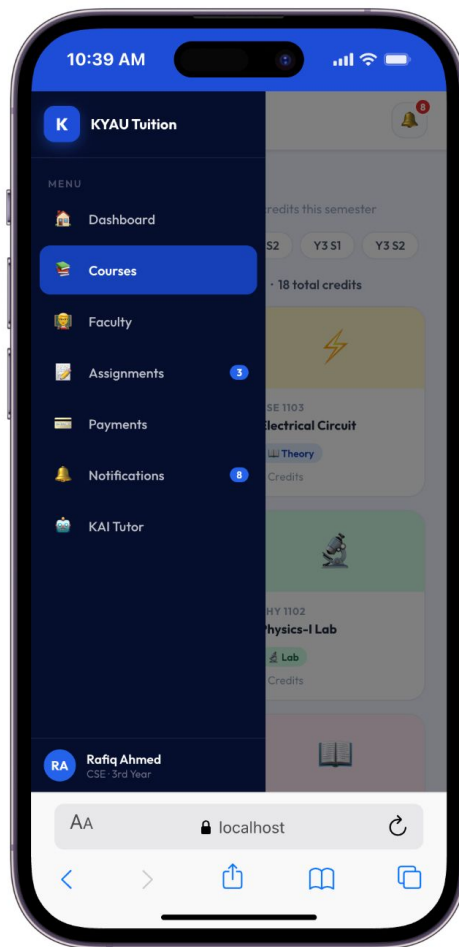


Figure 5.2: Student Dashboard — personalised welcome card, stat tiles (Courses, Pending Assignments, Today's Classes, Attendance %), today's timetable preview, current semester course cards with progress bars, and the latest departmental notice.

5.4 Courses Page

Figure 5.3 shows the eight-semester BSc CSE course browser.

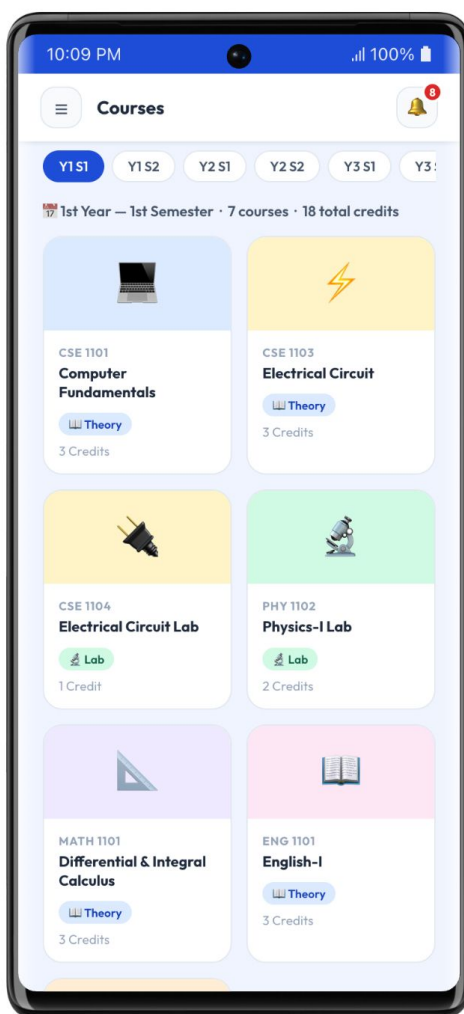


Figure 5.3: Courses module — semester tab selector (Semester 1 through 8); each tab lists the courses for that semester with course code, full name, credit count, and a Theory or Lab type badge.

5.5 Student Assignments

Figure 5.4 shows the Student Assignments page with tab-based status filtering.

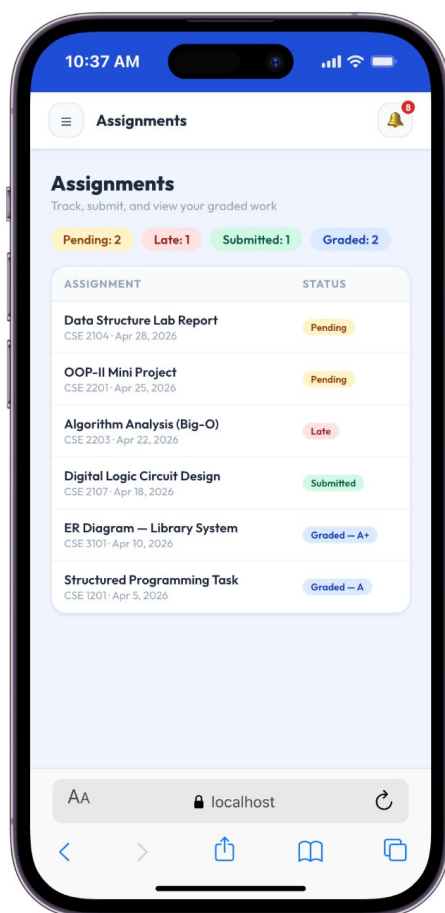


Figure 5.4: Student Assignments — all assignments listed with colour-coded status pills (Pending, Submitted, Late, Graded); tab bar for filtering by status; context-sensitive action buttons (Submit / View Submission / View Feedback).

5.6 Authentication and Role-Based Routing

The `useAuth` hook wraps a React context that holds the current user object. On login, the email suffix determines the role: `@teacher.kyau.edu.bd` loads the teacher portal and `@student.kyau.edu.bd` loads the student portal. The `Shell` component reads this role and selects the correct navigation array and page components.

5.7 KAI Chatbot Implementation

The `ChatPage` component manages a `messages` array in React state. On each send, the last 20 messages are sent to the Claude API with a role-specific system prompt. The response is rendered via a lightweight `fmt()` function that converts markdown into HTML. Figure 5.5 shows KAI responding to a student’s academic query.

```
const res = await fetch('https://api.anthropic.com/v1/messages', {
  method: 'POST',
  headers: { 'Content-Type': 'application/json' },
  body: JSON.stringify({
    model: 'claude-sonnet-4-20250514',
    max_tokens: 1000,
    system: system,    // role-aware university context prompt
    messages: history // last 20 conversation turns
  })
});
const data = await res.json();
return data?.content?.[0]?.text || 'Sorry, try again.';
```

Listing 5.2: Claude API Call in ChatPage.jsx

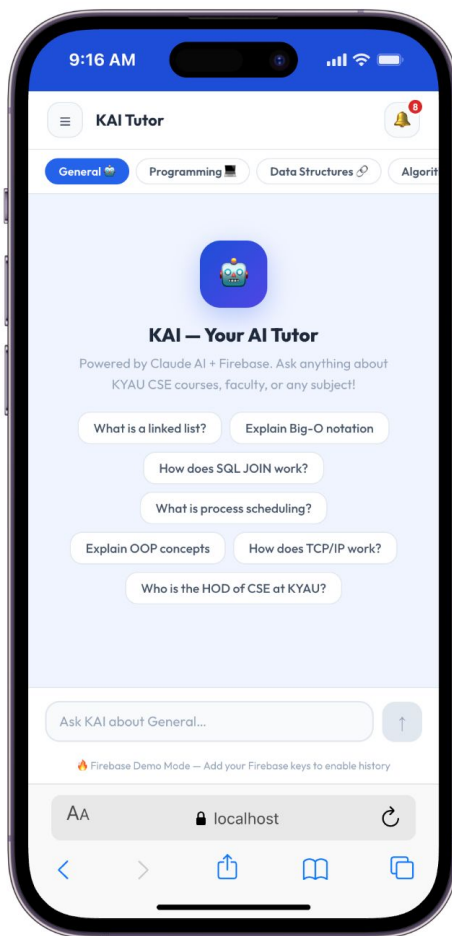


Figure 5.5: KAI AI Tutor (student mode) — subject chips at the top (Programming, Data Structures, Algorithms, etc.), quick-prompt suggestions on the welcome screen, and an active conversation showing KAI answering a CSE academic question with a structured, context-aware response.

5.8 Grading Logic

The `StudentGrades` component filters `STUDENT_SUBMISSIONS` for the logged-in user's graded entries, maps each letter grade to its grade point via a lookup object (`GP`), and computes the average GPA displayed to two decimal places.

5.9 Android APK Generation

```
npm run build
npm install @capacitor/core @capacitor/cli @capacitor/android
npx cap init "KYAU Tuition" "com.kyau.tuition" --web-dir dist
npx cap add android
```

```
npx cap copy  
npx cap open android # opens Android Studio  
# In Android Studio: Build > Generate Signed APK
```

Listing 5.3: Steps to Build Android APK

Chapter 6

Testing and Output

6.1 Testing Strategy

The application was tested through manual functional testing across three categories: component-level UI testing, role-based flow testing, and AI integration testing. A separate round of responsive testing was conducted by resizing the browser window and running the app on an Android emulator.

6.2 Test Cases and Results

Table 6.1: Functional Test Cases

#	Test Case	Expected Result	Status
1	Student login with valid credentials	Student portal loads with correct name and role	Pass
2	Teacher login with valid credentials	Teacher portal loads with teacher-themed UI	Pass
3	Login with invalid credentials	Error message displayed, no portal access	Pass
4	Student navigates to Courses	All 8 semester tabs render with correct course data	Pass
5	Student switches semester tab	Course list updates without page reload	Pass

#	Test Case	Expected Result	Status
6	Student views Assignments	All assignments shown with correct status pills	Pass
7	Student filters assignments by status	Only matching assignments displayed	Pass
8	Student views Grades	GPA computed correctly from graded submissions	Pass
9	Student views Timetable	Weekly schedule renders all 5 days	Pass
10	Student reads Notices	All notices displayed with type tags and dates	Pass
11	Teacher views Students list	All students rendered in card grid	Pass
12	Teacher searches student by name	Filtered results update in real time	Pass
13	Teacher views student profile	Profile card with GPA, attendance, contact info shown	Pass
14	Teacher creates assignment	Form fields validate and alert confirms posting	Pass
15	Teacher opens grade view	All submissions for selected assignment listed	Pass
16	Teacher enters and saves grade	Grade input captured and alert displayed	Pass
17	Teacher views schedule	Personal schedule filtered from full timetable	Pass
18	Teacher sends message to student	Message appears in chat with timestamp	Pass
19	KAI responds to student query	Context-aware academic reply returned from Claude	Pass
20	KAI responds to teacher query	Teaching-context reply returned from Claude	Pass
21	Notification badge updates	Bell icon badge reflects unread count	Pass

#	Test Case	Expected Result	Status
22	Sidebar collapses on mobile	Drawer overlay appears, content is accessible	Pass
23	Logout clears session	User is returned to login page	Pass

6.3 Sample Outputs

6.3.1 Student Dashboard

On logging in with `rafiq@student.kyau.edu.bd`, the student dashboard displays a personalised welcome card reading “Welcome back, Rafiq!” with the tag “Spring 2026 · BSc CSE · KYAU”. The stat tiles show 8 Courses, 3 Pending assignments, 2 classes today, and 87% Attendance (see Figure 5.2).

6.3.2 Course Browser

Selecting different semester tabs correctly loads the courses for each of the 8 semesters. For example, Semester 5 shows Data Communication and Computer Networks (CSE 3101, 3 credits), Database Management Systems (CSE 3103, 3 credits), and their lab counterparts (see Figure 5.3).

6.3.3 KAI Chatbot Interaction

When a student types “What is a linked list?”, KAI returns a structured explanation covering definition, node structure, time complexities, and a use-case example — all within 250 words as configured in the system prompt (see Figure 5.5).

6.3.4 Notification Bell and Dropdown

Figure 6.1 shows the notification bell badge displaying the unread count, and the dropdown preview that appears on click, listing recent notifications with a “View All” link.

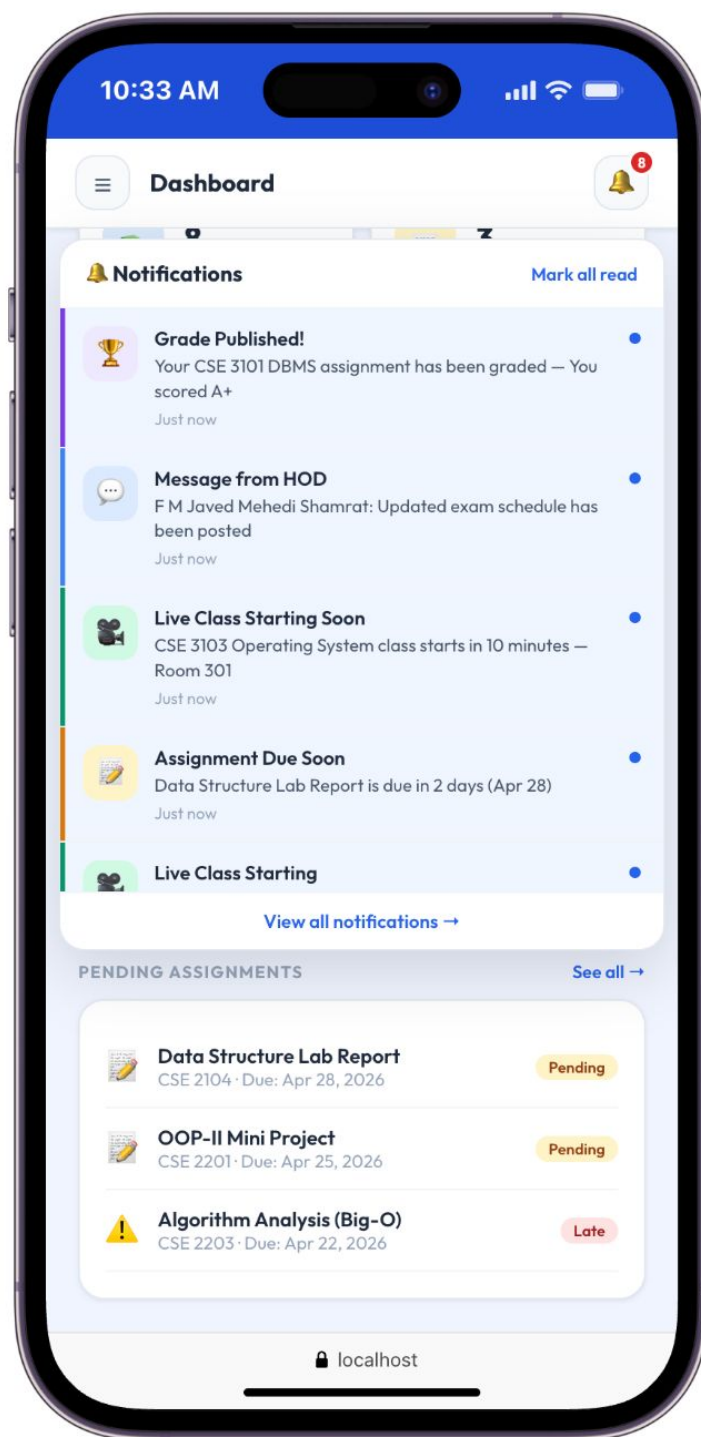


Figure 6.1: Notification bell badge showing the unread notification count, and the drop-down preview listing the latest notifications with timestamps and a “View All” shortcut link.

Chapter 7

Results and Discussion

7.1 Results

The KYAU Tuition App v2.0 was successfully developed and tested across all its planned features. The application achieved the following outcomes:

- A fully functional dual-role SPA was built with complete student and teacher portals, each with distinct navigation, theming, and feature sets.
- All eight semesters of the KYAU BSc CSE curriculum — containing over 40 courses with accurate codes, credit values, and type classifications — were embedded and made browsable through a tab interface.
- The assignment lifecycle (creation → submission → grading → result display) was implemented end-to-end for both portals.
- GPA computation was implemented using the standard 4.0-scale grade-point mapping.
- KAI was successfully integrated with the Anthropic Claude API and proved capable of answering both KYAU-specific queries and general CSE academic topics.
- The responsive layout was validated on both desktop and narrow mobile viewports.
- The Capacitor-based Android build process was validated to produce a functional APK through Android Studio.

7.2 Discussion

The project demonstrated the power of a component-based frontend architecture in building a feature-rich application without a dedicated backend server. React context hooks keep the code clean and free of prop-drilling even as the component tree grows.

The KAI integration proved to be the most impactful feature from a user experience perspective. Injecting faculty data, course lists, and user identity into every API call gives KAI contextual intelligence that general-purpose chatbots lack. The trade-off is that each conversation is stateless from the server's side; the app compensates by sending the last 20 messages as history.

The use of static data for most academic content makes the app fast and easy to deploy, but real-time teacher-to-student updates would require migrating to a fully Firestore-backed model for production. The dual-role colour theming — blue for students, teal-green for teachers — consistently helped testers orient themselves within the correct portal.

Chapter 8

Future Work

While the current version delivers a comprehensive feature set, several enhancements are planned for future releases:

1. **Full Firebase Firestore Integration:** Migrate all dynamic data to Firestore for real-time cross-device updates.
2. **Push Notifications:** Implement Firebase Cloud Messaging (FCM) for assignment, grade, and notice alerts on Android.
3. **File Upload for Assignment Submission:** Integrate Firebase Storage for PDF and image uploads.
4. **Attendance Management:** Teacher-side attendance marking with student history and below-75% alerts.
5. **Video Class Integration:** In-app Google Meet or Jitsi link generation from the schedule view.
6. **iOS Support:** Add Capacitor's iOS target for Apple App Store distribution.
7. **Admin Portal:** A third role for the Department Admin to manage timetables, faculty, and announcements.
8. **Offline Mode:** Service worker caching for courses, timetable, and notices on poor connections.
9. **Gradebook Analytics:** GPA trend charts and semester-by-semester progress visualisations.

10. **KAI Voice Mode:** Web Speech API integration for voice-based queries and spoken responses.

Chapter 9

Conclusion

The KYAU Tuition App v2.0 represents a meaningful step toward digitising academic management for the CSE department of Khwaja Yunus Ali University. We successfully designed and implemented a full-featured, dual-role platform that addresses real pain points — from fragmented assignment tracking and inaccessible timetables to the absence of on-demand academic support.

Building in React and Vite gave us a fast, maintainable codebase. Firebase provided the authentication and real-time communication backbone, while Capacitor made Android deployment possible from a single codebase. The KAI AI tutor — grounded in KYAU-specific institutional context via the Anthropic Claude API — was the standout feature and demonstrated how modern LLMs can be practically applied in education.

Throughout the project, the team gained hands-on experience in component-based design, context-driven state management, REST API integration, responsive UI development, and mobile deployment — skills directly relevant to the software industry. We are confident that the planned enhancements will evolve KYAU Tuition App into a production-ready system that genuinely improves daily academic life at KYAU, and we hope it serves as a strong foundation for future batches to build upon.

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Contribution Table

The following table describes the individual contributions of each team member to the KYAU Tuition App project.

Table 9.1: Team Member Contributions

Name	Student ID	Contribution
Shimanto Kumar (Team Leader)	0622220105101023	Led overall project planning and coordination. Designed and implemented the core architecture (App.jsx, Shell, role-based navigation). Developed the Student Dashboard, Teacher Dashboard, Courses, and Grades modules. Implemented <code>useAuth</code> hook and role-based routing. Managed Firebase integration and coordinated the Capacitor APK build. Wrote the Introduction, System Design, and Conclusion chapters of this report.

Name		Student ID	Contribution
Neyamul Islam		0622220105101005	Developed the KAI AI chatbot (<code>ChatPage.jsx</code>): Anthropic Claude API integration, system prompt design for both personas, conversation history management, and markdown-to-HTML rendering. Implemented the Teacher Assignments module. Designed the notification system (<code>useNotifications</code> hook), dropdown, and notifications page. Conducted testing for assignment and grading workflows. Contributed to the Implementation and Testing chapters.
Shaem Shakib	Hasan	0622220105101024	Designed and implemented Student Assignments, Student Timetable, Student Notices, Teacher Students, Teacher Schedule, and Teacher Messages modules. Developed the static data file (<code>kyauData.js</code>) covering all eight semesters, faculty, timetable, students, assignments, and messages. Handled all CSS styling — design tokens, responsive layout, role-based theming, progress bars, and status pill badges. Wrote the Related Work, Tools and Technologies, and Future Work chapters.

All members contributed equally to user testing, peer code review, and final report editing.